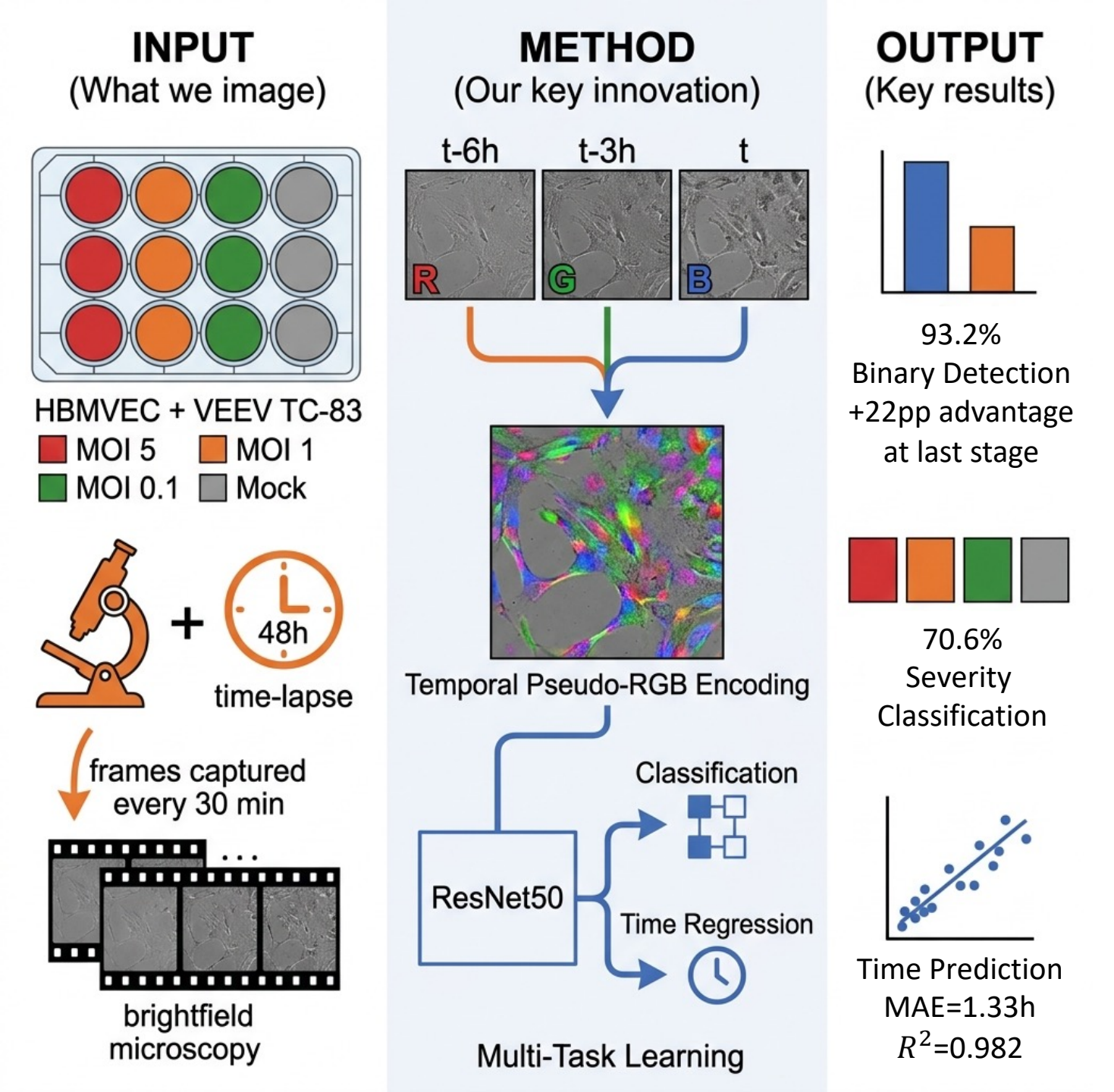
INPUT
(What we image)
HBMVEC + VEEV TC-83
MOI 5
MOI 1
MOI 0.1
Mock
48h
time-lapse
frames captured
every 30 min
brightfield
microscopy
METHOD
(Our key innovation)
t-6h
t-3h
t
R
G
B
Temporal Pseudo-RGB Encoding
ResNet50
Classification
Time Regression
Multi-Task Learning
OUTPUT
(Key results)
93.2%
Binary Detection
+22pp advantage
at last stage
70.6%
Severity
Classification
Time Prediction
MAE=1.33h
$R^2$=0.982